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Multicomponent magnetization of the Ordovician Age Younger Gabbros, Aberdeenshire, Scotland

C. M. Carmichael* *Department of Geophysics, University of Edinburgh, Edinburgh EH9 3JZ, Scotland*

K. M. Storetvedt *Department of Geophysics, University of Bergen, Bergen N-5014, Norway*

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Summary. A study of the magnetization of 12 block samples of the Huntly and Insch masses of the Younger Gabbros of Aberdeenshire, Scotland, has shown that these bodies have a complex multicomponent magnetization. Combined AF and thermal treatment has isolated both polarities of two nearly horizontal axes directed NNW–SSE and NNE–SSW although only the first of these axes is obvious in the NRM directions. Microscope examination, electron probe analysis and Curie point studies indicate the magnetization to be due mainly to magnetite and haematite needles exsolved within pyroxene crystals and haematite formed by oxidation of a few primary magnetite grains. It is inferred that the Younger Gabbros were magnetized between their Rb-Sr emplacement age of 489 ± 17 Ma and their K-Ar cooling age of 468 ± 4 Ma, by a combination of thermochemical and exsolution processes. The two axes of magnetization yield pole positions of 189.6° E, 27.8° N (dp = 4.7°, dm = 9.4°) and 151.6° E, 25.5° N (dp = 4.0°, dm = 7.9°). In view of the unsatisfactory scatter in British Ordovician pole positions it is suggested that these be assessed giving consideration to possible unresolved multicomponent magnetizations as well as tectonic relationships, precise time of magnetization and evidence that the field was dipolar in nature.

Introduction

Multicomponent magnetization of palaeomagnetic samples is being recognized as a rather frequent occurrence. It is quite commonly found in red bed sediments, in which the magnetic components associated with the red cementing matrix often lie along an axis oriented differently from that of the components carried by the detrital grains. Some recent representative examples of studies in which the different components have been separated by combining techniques of thermal demagnetization, acid leaching and specimen cutting are given

* Permanent address: Department of Geophysics, University of Western Ontario, London, Ontario N6A 5B7, Canada.

by Henry, Mank & Van der Voo (1977), Roy & Robertson (1978), Morris & Carmichael (1978) and Storetvedt & Carmichael (1979).

Though multicomponent magnetization has been detected less frequently in igneous rocks, examples of cases in which two or more components have been isolated have been given by Buchan & Dunlop (1976), Storetvedt *et al.* (1978), Palmer *et al.* (1979), Roy, Anderson & Lapointe (1979) and Piper (1979). Most of these cases are interpreted as resulting from partial remagnetization by various thermo-chemical mechanisms at some time after the original emplacement of the igneous body.

This paper is concerned with the magnetization of some samples from the Huntly, Insch and Boganclogh bodies of the Younger Gabbros of Aberdeenshire, an extensive system of Ordovician basic intrusives in north-east Scotland. The samples were collected for a palaeointensity study of the early Palaeozoic field following the publication of a palaeomagnetic investigation of the gabbros by Sallomy & Piper (1973) as part of the series entitled 'Palaeomagnetic studies in the British Caledonides'. During the investigation of the magnetization of the samples, which was quite unsuccessful as far as producing reliable estimates of the palaeointensity of the Ordovician field is concerned, the evidence for the multicomponent nature of the magnetization was obtained.

Geological setting

The Younger Gabbros of Aberdeenshire are part of an igneous complex in north-east Scotland that has undergone a number of metamorphic and tectonic episodes, most of which seem to be associated with Caledonian regional metamorphism. An extensive review of the literature on the area was given by Sallomy & Piper (1973). Detailed studies of the structure of the Belhelvie and Insch masses of the Gabbro have been reported by Boyd & Munro (1978) and Ashcroft & Munro (1978) respectively. There is persuasive evidence for the following picture. The Younger Gabbros post-date the maximum period of Caledonian metamorphism which strongly affected the more metamorphosed Older Gabbros. The various masses which now outcrop over an area of 2500 km^2 were either intruded as a single sill-like body up to 6 km in thickness or as several bodies of somewhat lesser thickness spaced closely together in time. Following crystallization there was extensive shearing and folding along axes with various local orientations so that the cumulate banding structure of the masses now forms a very complex pattern. The last of the folding episodes was on a broader scale along NNE–SSW axes associated with the Boyndie Syncline and the Buchan Anticline. All of these movements, however, seem to have taken place after crystallization of the masses but while they were still at relatively high temperatures, 500°C or greater.

The timing of these events is estimated from radiometric dates given by Pankhurst (1970, 1974). These consist of Rb-Sr whole rock isochrons that yield 489 ± 17 Ma for the Insch and Haddo House masses of the Younger Gabbros and 452 ± 4 Ma for the younger Kennethmont granite that cuts the Insch mass (other granites in the area give similar ages), as well as numerous K-Ar dates on micas within the gabbros that give a very well-grouped set of ages averaging to 468 ± 4 Ma. (Note, these ages have been recalculated using the 1976 IUGS constants, Steiger & Jager 1977.)

We can assume that the Rb-Sr age of the gabbros, 489 Ma, is the time of crystallization and that the K-Ar age of 468 Ma is the time of cooling, to 150°–200°C say, probably due to the gabbros being uplifted. The intrusion of the Kennethmont and other granites is a slightly later igneous event. This strongly suggests that the Younger Gabbros underwent a relatively long cooling interval of some 20 Myr during early Ordovician time. The magnetization of the gabbros should date from this interval. It was this evidence of the time of magnetization, the descriptions of the bodies as relatively unmetamorphosed, and the

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evidence of palaeomagnetic stability that prompted the collection of the samples for palaeointensity purposes.

Samples and results

A modest collection was deemed suitable for palaeointensity measures and thus 12 oriented block samples were collected, two samples from each of six locations. The grid references of the locations are given in Table 1. There were two in the Huntly mass, sample numbers H-1

Table 1. Location, treatment information and results for Younger Gabbro specimens.

Grid Ref.	No.	NRM D°	NRM I°	Treatment	Remarks	Fisher Means D°	Fisher Means I°	α_{95}°
	HUNTLY							
NJ 517408	H-1-1	338	+ 1	AF to 20 mT	non-repeatable, no grouping			
	H-1-2	70	-44	AF to 20 mT, thermal to 670°	3 steps 620°C to 660°C group(Fig 2)	202.1	+14.9	22
	H-1-3	268	-35	AF to 20 mT, thermal to 660°	poorly repeatable, no grouping			
	H-1-4	338	-31	2 year storage test	NRM repeat D:338, I:-29			
	H-1-5	128	-74	thermal to 640°C	no grouping			
	H-2-1	8	-42	AF to 10 mT	non-repeatable, no grouping			
	H-2-2	251	+71	thermal to 660°C	no grouping			
	H-2-3	343	+17	AF to 10 mT, thermal to 670°C	3 steps 620°C to 660°C group(Fig.2)	176.5	+ 8.6	25
	H-2-4	201	-48	thermal to 640°C	non-repeatable, no grouping			
NJ 518408	H-3-1	191	+55	AF to 70 mT	repeatable, little change			
	H-3-2	197	+59	AF to 80 mT, thermal to 690°C	7 steps 600°C to 680°C group(Fig.2)	185.6	+16.7	9
	H-3-3	202	+56	2 year storage test	NRM repeat D:206, I:+58			
	H-3-4	191	+55	2 year storage test	NRM repeat D:191, I:+54			
	H-3-5	192	+55	thermal to 685°C	no change to 500°C then no grouping			
	H-3-6	196	+58	thermal to 670°C	7 steps 590°C to 670°C group(Fig.2)	172.5	+17.4	14
	H-4-1	46	+13	AF to 20 mT, thermal to 660°C	non-repeatable, no grouping			
	H-4-2	118	- 9	2 year storage test	NRM repeat D:114, I:+1			
	H-4-3	100	+25	AF to 20 mT,D:320,I:+59 2 year storage test	repeat 20 mT D:321, I:+62			
	H-4-4	49	+25	thermal to 685°C	4 steps 620°C to 685°C group(Fig.2)	187.2	• 5.0	19
	INSCH							
NJ 626276	I-1-1	271	+55	AF to 30 mT	poorly repeatable, no grouping			
	I-1-2	352	-17	AF to 30 mT	3 steps 10 to 20 mT group	21.3	+ 3.9	10
	I-1-3	2	-38	AF to 30 mT	4 steps 12.5 to 20 mT group	16.8	- 4.5	6
	I-1-4	26	+45	AF to 20 mT, thermal to 550	9 steps 12.5 mT to 525° group	16.5	- 7.2	3
	I-1-5	61	-53	thermal to 650°C	4 steps 300°C to 550°C group(Fig.3)	26.7	- 6.9	6
	I-1-6	87	-20	thermal to -50°C	poorly grouped			
	I-2-1	105	-26	AF to 45 mT, thermal to 680°C	poor grouping, non-repeatable			
	I-2-2	33	-37	thermal to 680°C	non-repeatable			
	I-2-3	255	+70	AF to 50 mT	5 steps 10 mT to 20 mT group(Fig.5)	29.5	- 4.9	5
NJ 587264	I-3-1	155	- 3	AF to 100 mT	repeatable, all steps group	152.4	- 6.4	2
	I-3-2	146	- 3	AF to 30 mT, thermal to 650°C	all steps to 550° group	144.3	- 5.5	2
	I-3-3	151	- 3	2 year storage test	D:152, I:-4			
	I-3-4	155	+ 4	thermal to 660°C	repeatable, all steps group	157.3	- 2.8	5
	I-4-1	155	+ 4	AF to 30 mT, thermal to 650°	all steps to 550° group	157.6	+ 3.6	2
	I-4-2	153	0	2 year storage test	D:153, I:+1			
	I-4-3	158	+ 4	thermal to 640°	all steps to 640°C group	161.7	+ 5.1	9
	I-4-4	151	+ 2	thermal to 670	all steps to 670°C group(Fig.4)	154.4	+ 2.1	5
NJ 577246	I-5-1	216	+74	thermal to 590°C	no grouping			
	I-5-2	225	+67	AF to 250 mt	poorly repeatable			
	I-5-3	236	+69	thermal to 620°C	3 steps 575°C to 620°C group(Fig.4)	163.8	- 4.3	22
	I-5-4	229	+72	thermal to 590°C	no grouping (Fig.3)			
	I-5-5	256	+61	thermal to 620	poor grouping			
	I-6-1	350	- 9	AF to 50 mT, thermal to 670°C	see text (Fig.3)	349.6 200.4	-15.9 +11.8	3 15
	I-6-2	352	-12	thermal to 675°C	see text (Fig.4)	350.0	-12.4	3
	I-6-3	351	-11	thermal to 550°C	repeatable to 500°C, little change			
	I-6-4	352	- 9	AF to 50 mT, thermal to 550°C	all steps to 500°C group	355.8	-15.6	2
	BOGANCLOGH (INSCH)							
NJ 474284	I-7-1	221	+43	AF to 125 mT, thermal to 640°C	3 steps 600°C to 640°C group(Fig.4)	167.6	+29.7	19
	I-7-2	216	+44	thermal to 790°C	4 steps 680 to 790 group (Fig.3)	178.0	+16.8	5
	I-7-3	217	+43	thermal to 675°C	no grouping			
	I-7-4	213	+47	thermal to 790°C	poor grouping			
	I-7-5	216	+44	thermal to 690°C	no grouping (all I-7 specimens unchanged to 550°C)			
	I-8-1	223	+41	AF to 50 mT, thermal to 660°C	3 steps 635°C to 660°C group(Fig.4)	148.9	+ 3.7	27
	I-8-2	221	+40	thermal to 685°C	3 steps 675°C to 685°C group	176.6	+18.6	46
	I-8-3	217	+44	thermal to 690°C	no grouping			
	I-8-4	214	+43	thermal to 650°C	no grouping			

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to H-4, three in the Insch mass, sample numbers I-1 to I-6 and one in the Boganclogh mass which is recognized as an extension of the Insch body and thus these samples were numbered I-7 and I-8. Comparing grid references with those of Sallomy & Piper (1973) it appears that our samples H-1 and H-2, and I-7 and I-8 came from their sites GH-5 and GC-3 respectively. From each block samples several specimens in the form of 2 cm right cylinders were drilled in the laboratory. These are numbered with an additional digit as H-1-1, H-1-2, etc.

Specimens from each sample were treated by alternating field (AF), and thermal demagnetization with the thermal treatment being applied in some cases after AF treatment and, in others, to untreated specimens. The AF demagnetizer has a three-axis tumbler, ambient field cancellation by a double Helmholtz coil system and Inductrol control of the AF field. The thermal demagnetizer has a non-inductively wound Pt heating coil with a Lindberg SB programmable control unit, six concentric cylindrical nested mu metal shields to provide an ambient field of about 1 nT, and provision for slow rotation of the specimens about one axis to assist in randomization of any moments that might be acquired during cooling.

Treatment in AF fields in steps of 2.5, 5 and 10 mT up to a maximum of 125 mT peak divided the samples into two categories, those that have some soft magnetic components (soft) and those that do not (hard). For the soft category, the individual specimens of the sample have very scattered, almost completely random, NRM directions and, upon AF treatment, the direction and intensity after each step is often scattered and not repeatable. The hard category has very good agreement between the NRM directions, little change in direction with AF treatment and good repeatability of individual steps, in some cases, up to the maximum of 125 mT. Both categories show evidence of quite hard magnetic components with median destructive fields in the range 20–50 mT. In the soft category samples the magnetization of these components is masked by the soft viscous components that are not randomized by the AF treatment, while in the hard category samples there is no such masking. The two categories are illustrated in Fig. 1. In most cases samples from the same

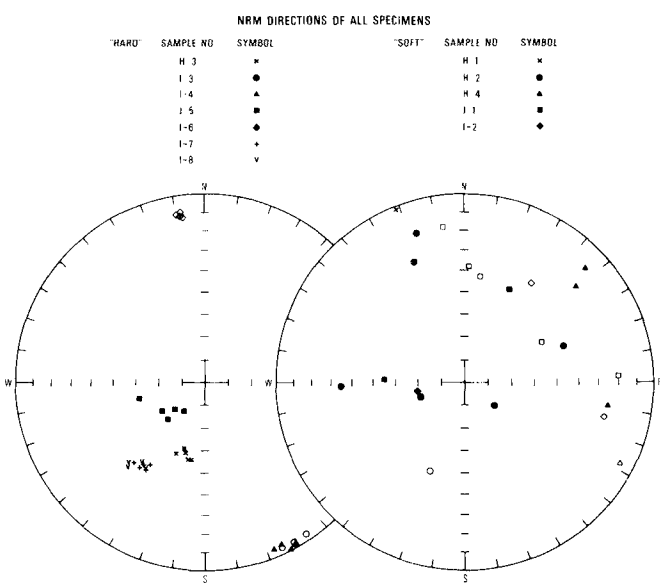


Figure 1. NRM directions of all specimens from the Younger Gabbro samples. Details of the samples are given in Table 1. In this and all other figures solid symbols indicate downward vectors and open or circled symbols indicate upward vectors.

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location fall in the same category but H-3 and H-4 from the same location do not. For the hard category, samples from the same location agree well in the case of I-3 and I-4, and I-7 and I-8 but not in the case of I-5 and I-6. Even for this category, however, it is disturbing to find that the directions, though very stable, are quite different and tend to remain so on AF treatment. There is the cluster of directions SSW and downward found by Blundell & Read (1958) and Sallomy & Piper (1973) but what is more interesting is the presence of both polarities of an almost horizontal axis of magnetization aligned in an NNW–SSE direction. Even so it is not a very promising palaeomagnetic collection on the basis of AF treatment.

A total of 38 specimens of the 12 samples were subjected to thermal demagnetization using closely spaced steps, repeated steps and repeated measurement in the higher temperature range. Problems with viscous components picked up upon transfer from the furnace to the magnetometer were encountered frequently but it was usually possible to remove these by spinning the sample in field-free space within the mu metal shields of the Schonstedt SSM-1 magnetometer. Decay of the viscous components was detected by within sample convergence of directions and intensity on repeated measurement. Some of the specimens, including all of the ones from sample I-2, were so viscous that repeat measurements after each thermal treatment step did not converge but satisfactory behaviour for some specimens was found for all other samples, even those that were unstable under AF treatment. In spite of the viscous component problems, thermal demagnetization was more successful in isolating a coherent magnetization direction than was AF treatment.

Figs 2–5 give a broad coverage of the results obtained. Within the magnetite blocking temperature region ($T < 580^{\circ}$C), there are two types of behaviour. Some specimens, including those which showed high stability on AF treatment, remain in their initial NRM position while others exhibit a quite complex swinging behaviour indicating an interplay between normal and reverse components, see H-1-2, H-2-3, H-4-4 of Fig. 2; I-5-4 of Fig. 3; I-2-3 of Fig. 5. Most of these were in the soft AF category but a part of their apparent unstable behaviour under AF treatment may also have been due to such an interplay between components for northerly and southerly shallow directions showed up among the AF steps of these samples more frequently than would be expected on a random distribution. In some of the examples, the directional data below *c.* 450°C have been left out in order to improve the clarity of the diagrams. Even some of the scattered but stable magnetic remanences are likely to represent vector resultants of normal and reverse components. Vector subtraction, applied to various vectors along directional paths, has not given consistent estimates of the erased vector. Only in cases where the directions swing rapidly from one polarity to the other are there reasons to believe that only one component is being removed. The direction of the subtracted vector in such cases is illustrated for specimens H-1-2 (Fig. 2), I-1-5 and I-1-6 (Fig. 3).

At $T > 600^{\circ}$C, the remanence distribution becomes much simpler. Specimens from all samples, except I-2, now attain a shallow magnetization and the declination is commonly directed SSE. This high temperature component is basically associated with haematite but in some of the specimens the high temperature direction has been followed up to 790°C, i.e. about 100°C above the Curie temperature of haematite (Fig. 3, I-7-2).

All of the specimens, except those from sample I-3, have multicomponents of magnetization with the most common component being directed SSE and nearly horizontal. Specimens from I-3 shown in Figs 1 and 3 seem to contain only this single flat-lying SSE component for their directions remain tightly grouped around their NRM direction under either AF or thermal treatment. The axis of magnetization defined by this direction is referred to as the *A*-axis. Some specimens in the collection, however, have components along another flat-lying axis directed SSW and referred to as the *B*-axis. Specimens H-1-2 in Fig. 2 and I-2-3 in Fig. 5 are good examples. In the case of I-2-3, the northerly polarity of the axis

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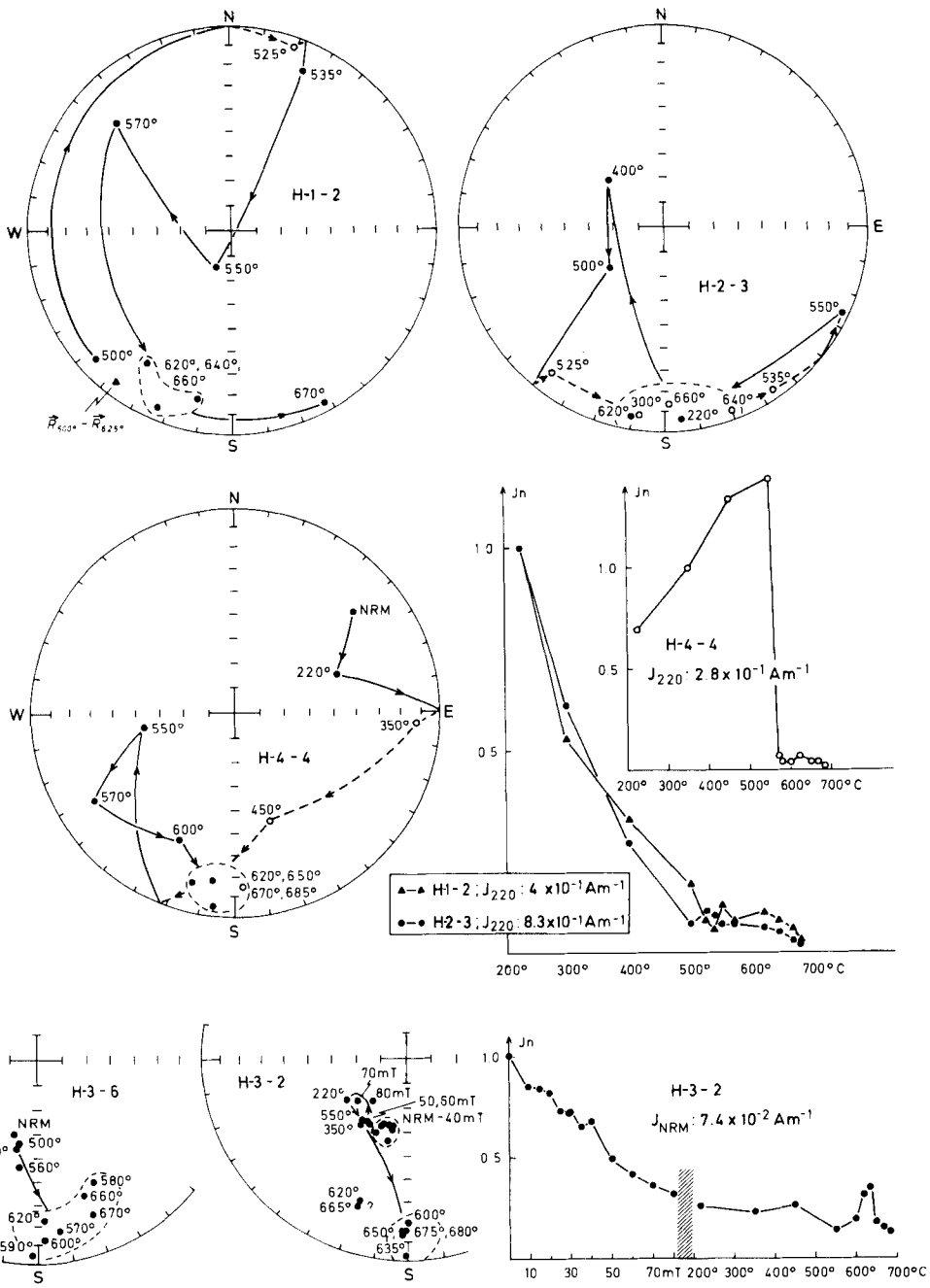


Figure 2. Demagnetization data for some specimens from the Huntly mass. Thermal demagnetization steps are indicated by the temperature in °C while alternating field demagnetization steps are indicated by the value of the peak field in mT. Specimen H-1-2 contains both polarities of the *B*-axis while H-2-3, H-3-2 and 6 and H-4-4 record the southerly *A*-axis vector, mainly in the haematite minerals. Some portion of the low Curie temperature moment in H-1-2 and H-2-3 is probably due to pyrrhotite.

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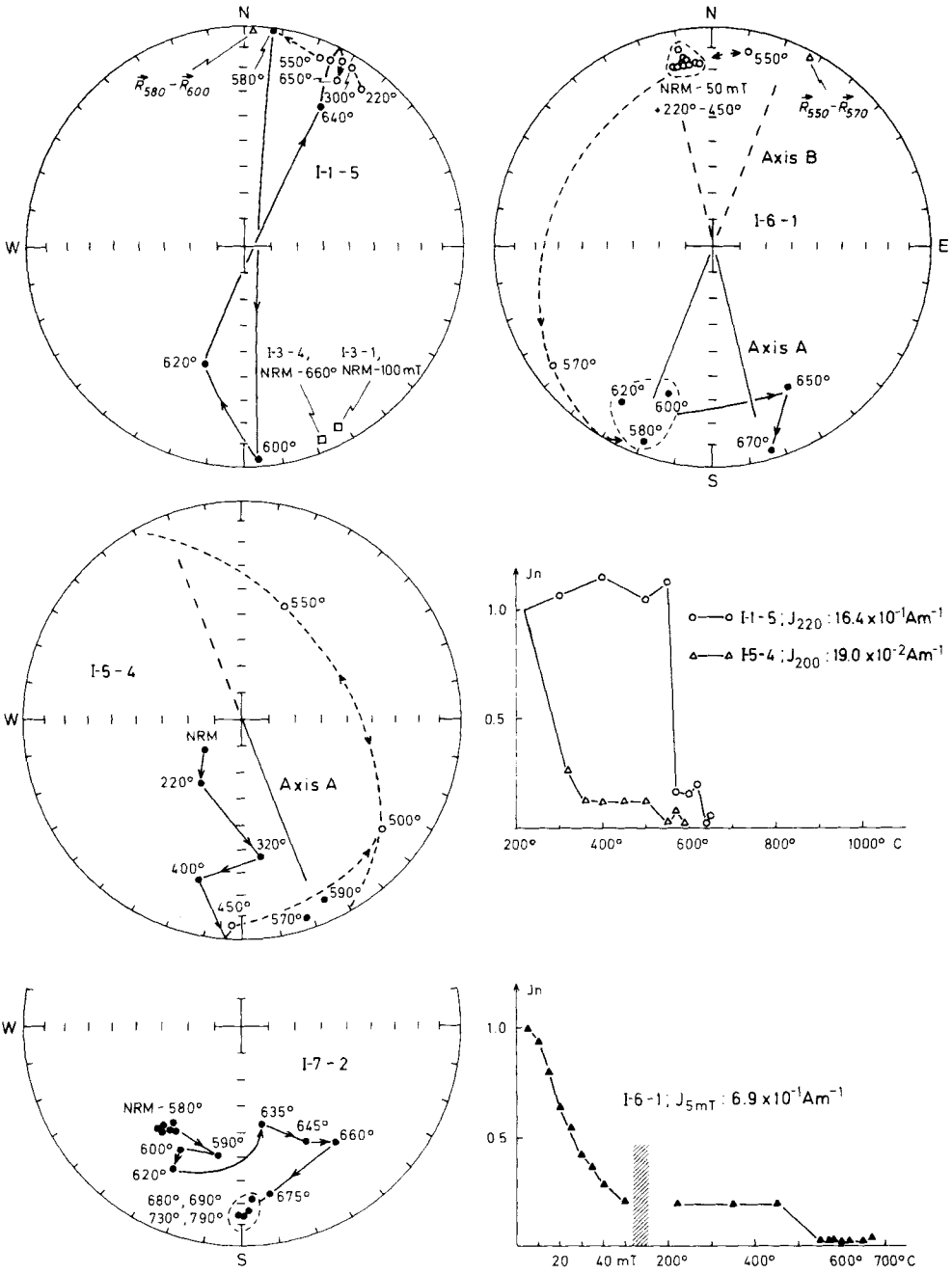


Figure 3. Demagnetization data for some specimens from the Insch mass with symbols as in Fig. 2. Specimens I-3-1 and I-3-4 in the upper left plot show sample I-3 to contain only the southerly *A* axis magnetization. Specimen I-6-1 in the upper right plot shows evidence of both axes while I-7-2 in the lower left plot shows a consistent remanence up to 790°C. Specimen I-5-4 has only two end points defining the southerly *A*-axis so it is not used in calculating the I-5 sample direction.

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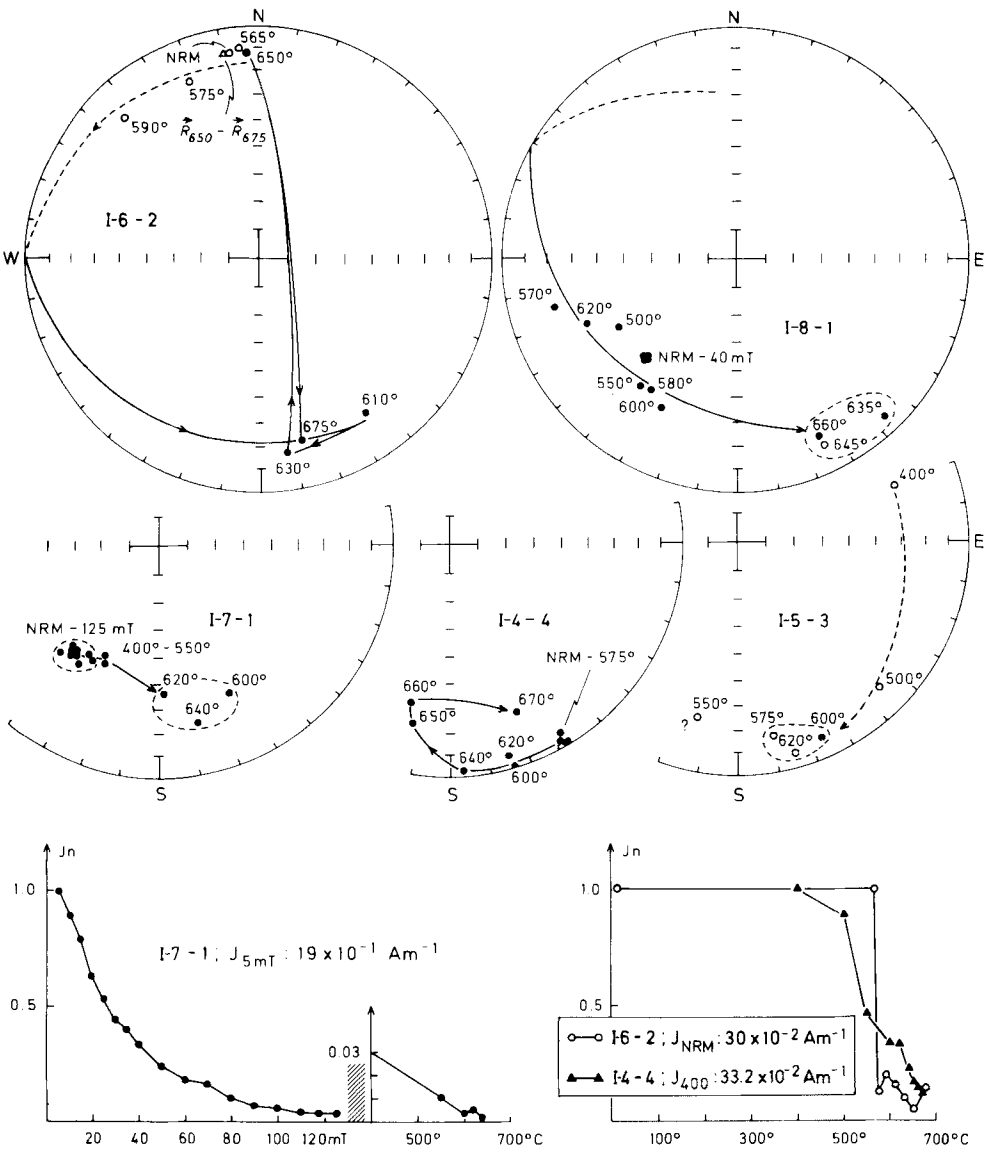


Figure 4. Demagnetization data for additional specimens from the Insch mass with symbols as in Fig. 2. Specimen I-8-1 thermal demagnetization points are spread along a great circle path suggesting unresolved normal and reversed *A*-axis magnetization except for the highest temperature points.

is isolated very well by moderate levels of AF treatment while the southerly polarity is very poorly defined by higher levels. Both the *A*- and *B*-axes are defined by both polarities of magnetization with usually one axis dominating in any one specimen, but specimen I-6-1 in Fig. 3 shows evidence of both polarities of both axes.

The palaeomagnetic directions obtained for the individual specimens are listed in Table 1. The majority of the Fisher mean directions listed represent stable end point directions calculated as the mean of at least three closely grouped successive points at the end of the repeatably measurable range of treatment steps. An example of a specimen that does not reach an end point is number I-5-4 shown in Fig. 3. While it is clearly in the southerly *A*-axis grouping

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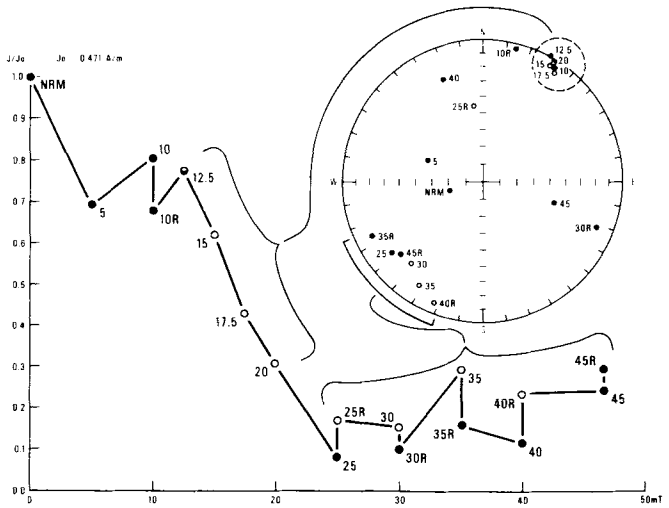


Figure 5. AF demagnetization data for specimen I-2-3. The northerly direction of the *B*-axis is well-defined at moderate fields and the southerly direction is poorly defined at higher fields. The letter R after a symbol indicates a repeat of the same field treatment.

at the 570°C and 590°C heating steps, it became too weak to measure at higher temperatures and our arbitrary criterion of a minimum of three grouped points could not be obtained. The values for I-6-1 and I-6-2 are exceptions. Specimen I-6-1 is the only one that is represented by two palaeomagnetic directions: the NRM–50 mT and the 220° 450°C directions have been combined with the 650°–670°C directions, and the 550° direction plus the vector subtracted component between 550°C and 570°C have been combined with the 580°–620°C group. Specimen I-6-2 in Fig. 4 shows both polarities of the *A*-axis magnetization. While it does not matter much how the data are combined, the direction for the sample was calculated by combining the thermal treatment steps up to 565°C with the vector subtracted between the 650°C and 675°C steps.

Table 1 gives the grid points of the sampling locations, the NRM directions of all specimens, the treatment applied, and the mean directions calculated from the results of three or more successive treatment steps that group together and are judged to define a stable end point. These mean directions for each specimen have been combined to obtain a sample mean and these are plotted in Fig. 8 with the northerly directions reversed and plotted as southerly. Even though samples collected at the same location give similar results in some cases, they do not in others and so each sample is treated as being independent. Those points that are interpreted as defining the *A*- and *B*-axes are indicated. Combining these points yields a mean for the *A*-axis of *D*: 169.1°, *I*: +8.4°, $\alpha_{95} = 9°$ and for the *B*-axis, *D*: 203.1°, *I*: +8.8°, $\alpha_{95} = 8°$. The corresponding pole position for the *A*-axis is 189.6° E, 27.8° N, $dp = 4.7°$, $dm = 9.4°$ and for the *B*-axis is 151.6° E, 25.5° N, $dp = 4.0°$, $dm = 7.9°$.

Magnetic mineralogy

The identity of the magnetic minerals within the gabbros has been inferred from thermal demagnetization data, Curie balance curves, microscope examination of polished thin sections and electron microprobe studies of some of the opaque oxide grains.

Microscopic examinations of polished thin sections of samples show them to be quite variable in terms of the amount, mode of occurrence and degree of alteration of their opaque minerals. In decreasing order of abundance these minerals are: ilmenite in the form

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of large grains up to 0.1 cm across, various sizes of iron sulphides both pyrite and pyrrhotite, and small needles, bars and blebs of magnetite, haematite and ilmenite exsolved within silicate grains. The silicate hosts for these small particles are mostly ortho- and clino-pyroxenes and biotite.

All samples have some degree of alteration of the opaque oxides but in general the alteration is noticeably less in the Huntly samples as compared to those from Insch. Photomicrograph (A) in Plate 1 shows a large grain of ilmenite under partly polarized light. It exhibits severe cracking, bending and complex twinning that is indicative of modification under high shearing stress but without chemical alteration. In most samples there is also evidence of considerable chemical alteration under oxidizing conditions. Ilmenite grains show alteration to rutile and minor haematite ranging in extent from none to almost complete conversion. Very little haematite seems to result from this alteration, however, since the products of the alteration are mostly rutile and translucent silicate. An example of this alteration is shown in Plate 1(D, E). In one case the silicate was identified by microprobe as sphene.

Only a few primary magnetite grains have been observed and most of these are almost completely oxidized to haematite as in Plate 1(B, C). Some of the magnetites show extensive fine-scale exsolution of spinel and ulvospinel which has divided the residual magnetite into small volumes on a scale of microns, as is shown in Plate 1 (F, G).

In all samples there are silicate grains, usually ortho- or clino-pyroxenes, within which needles, bars or plates of magnetite have exsolved (Plate 1, H). Strong shape anisotropy and small size make this form of magnetite very hard magnetically. The alignment of the long axis of the needles within the silicate grains will make the magnetization of each grain quite anisotropic, but because of the random orientation of the grains within the rock the magnetization of the rock is inferred to be isotropic. Etching with HF acid and microprobe analysis of some of the larger bars have shown that in addition to magnetite needles there are some that are ilmenite and some haematite. There is also some haematite that has formed in cracks and along grain boundaries. Alteration of some of the sulphides has produced haematite and a grey iron oxide that may be goethite as is shown in Plate 1(I).

Thermal demagnetization curves produced by plotting the magnetic moment measured at room temperature, after successive steps of heating and cooling in field-free space, are shown for representative specimens in Figs 2, 3, 4 and 6. Since changes in direction, also shown in these figures, indicate that components with widely different orientations are being removed the demagnetization curves are rather irregular. They do, however, indicate that a significant proportion of the magnetization is carried by minerals with unblocking tempera-

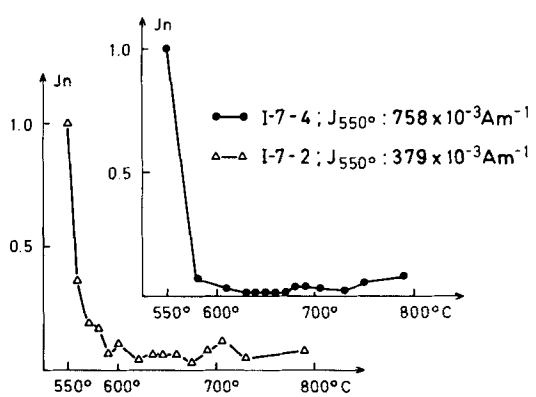


Figure 6. Thermal demagnetization data for two specimens from sample I-7 in which a coherent remanence persisted after heated to 790°C. Directional data for I-7-2 are shown in Fig. 3.

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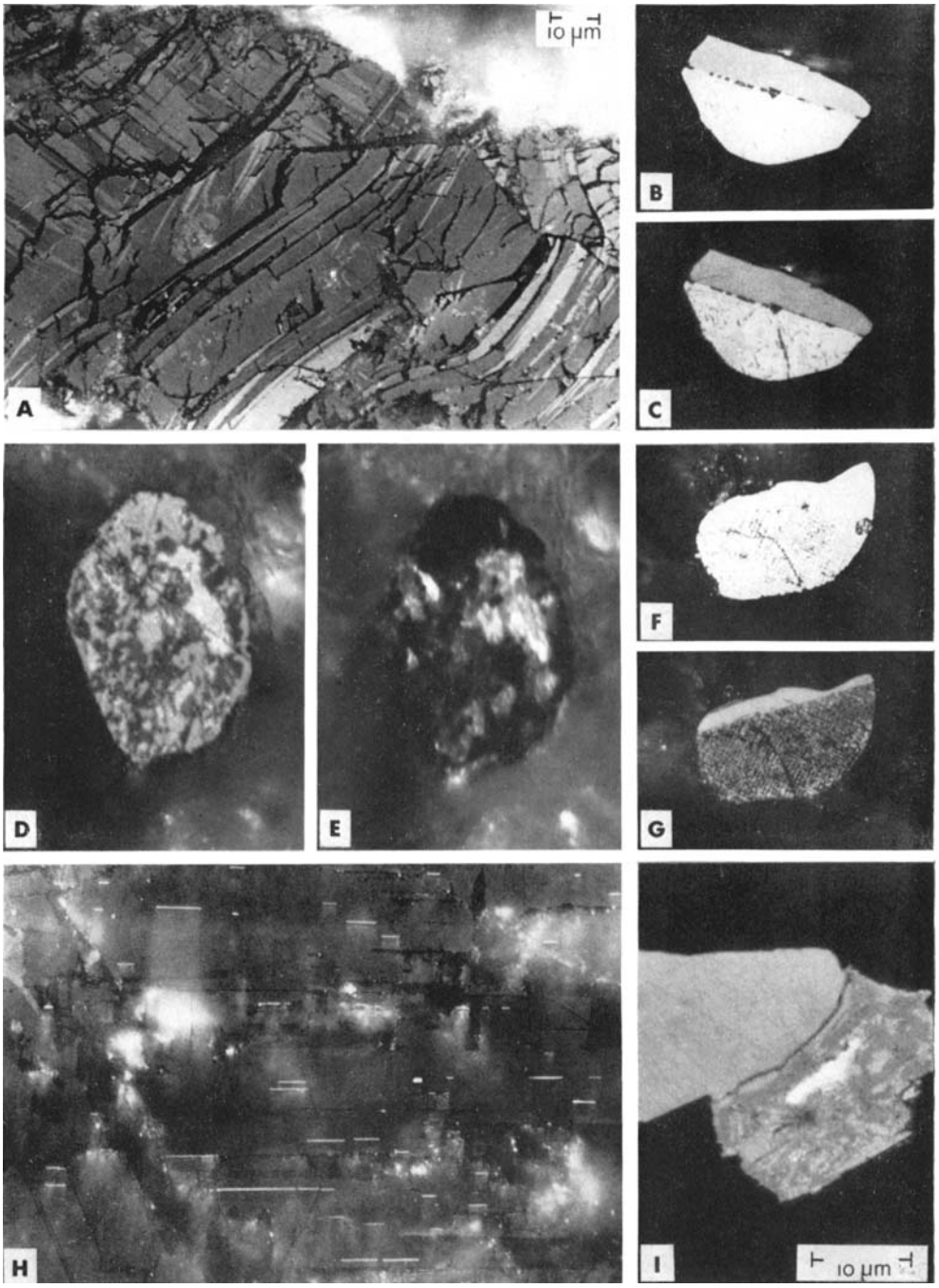


Plate 1. Photomicrographs of opaque grains in polished thin sections of Younger Gabbro samples: (A), large ilmenite from sample I-1 under partly polarized light showing bending, cracking and pressure twinning under high shearing stress; (D) and (E), small partly altered ilmenite grain from sample I-6. The residual ilmenite shows grey in (D) and black in (E) (under polarized light). Bright areas are rutile and dark grey areas are a silicate, possibly sphene. (B) and (C) and (F) and (G) from sample I-3 are grains of altered magnetite, each having a bar of ilmenite along the top. (C) and (G) are after etching with HF acid which turns magnetite black. The grain in (B) and (C) is almost completely converted to haematite which remains white on etching. The grain in (F) and (G) has considerable magnetite divided on a very small scale by spinel (dark in F) and ulvospinel (light in G) that have exsolved along the 100 planes. In (H) is shown a pyroxene grain from sample H-3 in which magnetite needles (white lines) have exsolved. (I) shows an altered pyrrhotite from sample I-7. The pyrrhotite has been converted to haematite, and an iron oxide that may be goethite, which show as zoned shades of grey around the residual pyrrhotite (white). The uniform grey area is an unaltered ilmenite with polishing scratches on it.

[*facing page* 208]

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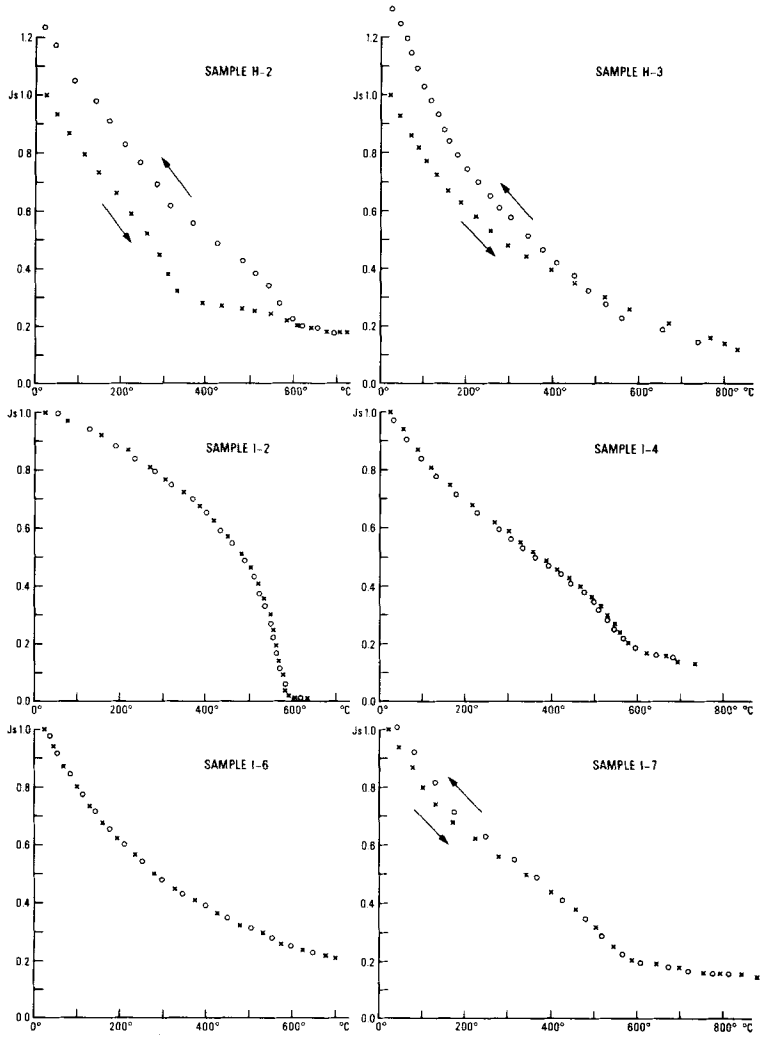


Figure 7. Curie balance curves measured in a field of 0.8 T. See text for details.

tures in the 600–700°C range and with several samples, measurable components with consistent directions exist up to at least 800°C as is shown in Fig. 6 for two specimens from I-7 that were treated in separate sequences of heating steps. In these cases, the remanent direction is stable and cannot be attributed to an artifact of cooling for the specimens were rotated during cooling and the remanence makes a large angle with the rotational axis. Each of the curves shows similar changes in intensity as well as a consistent direction.

Possible sources of this high-temperature magnetic moment could be cobalt-bearing oxides or the more unlikely metallic alloys of Fe, Co and Ni. Extensive optical examination and electron probe analyses of opaque grains failed to detect any such components. These need be present in only very small quantities and may be in very small grains, so we can neither say that such components are not present nor offer any alternative explanation for the high temperature magnetization.

The Curie balance curves for H-2 and H-3 of Fig. 7 are typical of samples from the Huntly mass. The cooling portion of the curve lies above the heating portion and is due to an

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increase in the amount of components with Curie points in the magnetite Curie point range. Such behaviour can be due to unmixing of titanium-rich titanomagnetites or by reduction of maghaemite but neither of these minerals have been observed in the Huntly samples. It is possible that the new magnetite is due to additional exsolution of needles in the silicates. In contrast to the curves of the Huntly samples, those for the Insch samples in Fig. 7 have the heating and cooling portions of their curves very close together indicating little change in the magnetic minerals upon heating. Change in the magnetic minerals upon heating does not correlate with the hard or soft category of Fig. 1: H-2 and I-2 are soft and the rest are hard.

The curve for sample I-2 is different from the rest in that it has a much greater proportion of its induced magnetization due to what is presumed to be magnetite because of the Curie temperature in the 560°–580°C range. In addition the magnitude of the induced moment for the sample I-2 was more than ten times as great as for the other samples in Fig. 7 although the arbitrary scale used does not show this. The remanent intensity of specimens from this sample, however, are not greater than for other samples, being about 0.5 $A m^{-1}$. This suggests that most of the magnetite contributing to the induced moment is so soft that it carries little remanence. A possibility for the source of this magnetite would be very fine exsolved needles on the edge of the super-paramagnetic size range that could receive an induced moment but no remanent magnetization.

The Curie balance curves of the other samples show that roughly one-fifth of the room temperature moment persists to temperatures well above the haematite temperature range. This tends to support the finding of a high-temperature remanence in sample I-7 described above but since these are Curie balance curves measured in a field of 0.8 T the effect could be due to paramagnetic minerals. However, paramagnetic effects as large as this have not been encountered with this instrument before.

Discussion

The palaeomagnetic directions determined by Sallomy & Piper (1973) for the Huntly and Insch masses are D: 177°, I: +53°, $\alpha_{95} = 23°$ and D: 149°, I: +26°, $\alpha_{95} = 14°$ respectively. The steeper inclinations of these directions, as compared to our results, seem to be due to a different choice of demagnetization treatment in selecting the data. Sallomy & Piper used data obtained by treatment in alternating fields of roughly 30–50 mT or after thermal demagnetization at temperatures in the 300°–550°C range. We have chosen to use data obtained by different combinations of demagnetization treatment that in general involved higher alternating fields and higher temperatures. The reason for this choice is that from our study we have inferred that treatment in lower fields and at lower temperatures does not resolve the multicomponent magnetization of these rocks.

We have concluded, as was suggested by Sallomy & Piper, that the hard magnetic components are mainly due to the fine needles and particles of magnetite and haematite exsolved within the various silicates. It seems probable that these needles and particles separated, and some were oxidized during the period of slow cooling inferred from the mineralogy and age data. They could thus be magnetized by a combination of exsolution, thermal and chemical magnetization processes and could preserve both normal and reverse orientations of the palaeomagnetic field. Because of similar size and shape the needles have similar coercive force spectra so it is not surprising that alternating field treatment is not very effective in resolving the components of the magnetization. The greater success of thermal treatment in resolving the components of the magnetization is interpreted as being due to the range of Curie points of the needles and particles produced by oxidation of the magnetite to haematite.

With rocks in which multicomponents of magnetization have been recognized, great care

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must be exercised in distinguishing separate components from the resultant of two or more components. Where the separate components are preserved in grains with identical coercive force of Curie point spectra the resultant direction will not change as the sample is demagnetized. We consider this to be the explanation for the apparent stability of the generally SSW and downward magnetizations of samples H-3, I-7 and I-8 (Figs 1, 2, 3 and 4) under AF demagnetization and thermal demagnetization below about 550°C. It is inferred that the various multicomponents of the magnetization being removed by these procedures all reside in fine needles of magnetite that are magnetically hard due to small size and large shape anisotropy and have essentially identical magnetic properties. The different sense of their magnetizations is interpreted as being due to different times of exsolution when the field was oppositely directed. It is of course possible that part of the tendency of the resultant magnetization of these different samples to group roughly is caused by the presence of a magnetization acquired over a short interval of needle exsolution during a reversal.

In any case, given the collection shown in Fig. 1, ordinary palaeomagnetic techniques would produce a poorly defined southerly and downward mean direction similar to those found by Blundell & Read (1958) and Sallomy & Piper (1973). All the soft category samples in Fig. 1 would be eliminated as unstable on the basis of within sample divergence. Of the hard category samples, I-5 and I-6 that come from the same location could be eliminated on the basis of within site divergence. Treating the remaining samples by AF demagnetization or thermal demagnetization at temperatures below 600°C would give the southerly downward mean direction.

Conclusions

We consider the magnetization of the Younger Gabbros of Aberdeenshire to be multicomponent in nature. We infer from our data that the bulk of the magnetic moment is carried by secondary grains of magnetite and haematite, some of which have exsolved within silicates. The observed alteration has involved high pressures and high temperatures as well as exsolution and oxidation at temperatures that were probably below the Curie temperatures of the oxides produced. While some of the magnetization may be simple TRM, most of it is probably a combination of chemical, partial thermoremanent, and crystallization magnetizations.

The major components of the multicomponent magnetization have been produced in both polarities of quite shallow fields directed NNW–SSE and NNE–SSW. It is not possible to distinguish the relative ages of these field directions, or to determine whether they represent two periods when the field axis was stationary or two episodes of conditions favourable for magnetization processes during an extended period of a steadily moving field axis.

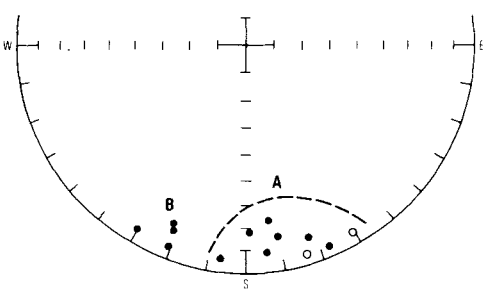


Figure 8. Mean directions of the specimens from each block sample are all plotted as southerly vectors. Those defining the *A*- and *B*-axes are indicated.

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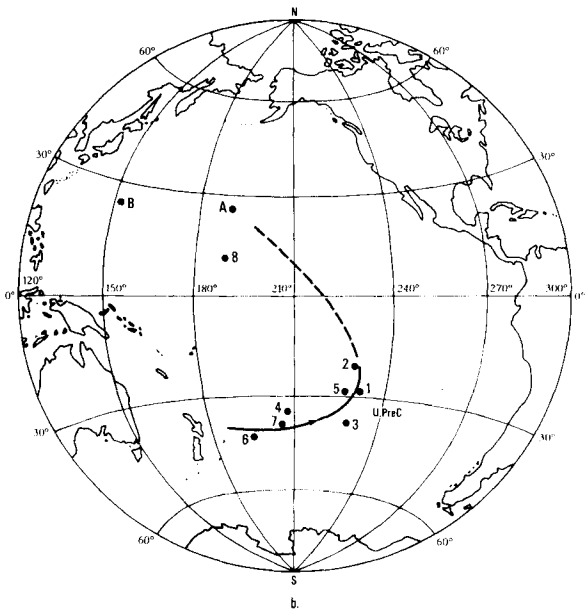


Figure 9. The virtual geomagnetic poles defined by the *A*- and *B*-axes along with some late Precambrian poles from NW Europe: 1 = Storetvedt & Gidskehaug (1968); 2–4 = Poorter (1972); 5 = Stewart & Irving (1974); 6 = Murthy & Deutsch (1976); 7 = Murthy & Deutsch (1974); 8 = Younger Gabbros (Sallomy & Piper 1973).

The timing of the magnetization of the Younger Gabbros is assumed to be the period between the Rb-Sr and K-Ar ages, i.e. between 489 ± 17 and 468 ± 4 Ma. It is also assumed that the K-Ar age is the time of cooling to the argon closure temperature and since there is remarkably little scatter in the individual K-Ar ages the uplift to produce cooling must have been a relatively rapid geological event. There is uncertainty about the argon closure temperature and the temperature below which magnetite needles could exsolve in silicates but we consider 150–200°C to be a reasonable minimum temperature for both. Under certain conditions, in submarine basalts for instance, magnetite can oxidize through maghaemite to haematite at lower temperatures than these but the magnetite needles in the gabbros are protected within the silicate grains and also do not show alterations similar to those observed in the submarine basalts. We consider it unlikely that significant stable magnetization was acquired after the time given by the 468 ± 4 Ma K-Ar age.

We are satisfied that the *A*- and *B*-axes directions and the poles they represent (shown in Fig. 9) define a segment of the polar wander path for this part of the Ordovician period. These poles differ from those summarized in Briden & Morris (1973) in that they fall about 15° farther north due to their shallower magnetization. With the growing recognition of the prevalence of multicomponent magnetization of igneous rocks it is suggested that a number of earlier studies might be re-examined. Some suites in which there is evidence similar to that of the Younger Gabbros for multicomponents that would also tend to produce a too steep magnetization would be the Connemara Gabbros of Ireland (Morris *et al.* 1973), the High Ireby and Carrock Fell groups of the Lake District (Briden & Morris 1973) and the Builth Volcanic Series of Wales (Piper & Briden 1973).

In an extensive study of the Welsh Borderlands, Piper (1978) has presented a number of Ordovician–Silurian poles. Three of the oldest of these, the tuffs of Cardonian age (about 450 Ma) and the Hendre and Blodwell units of the Berwyn Hills, yield poles similar to the

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Younger Gabbros although they are inferred to be some 20 Myr younger. The Foyers plutonic complex in Inverness-shire studied by Kneen (1973) contains similar poles with a similar spread in longitude. The age of the complex is given by Dewey & Pankhurst (1970) as between 490 and 410 Ma. Based on the Younger Gabbro data it would appear that the time of magnetization of the Foyers complex would be the early part of this age span.

Other units of Ordovician or presumed Ordovician age are magnetized close to the vertical and thus yield quite different pole positions (Thomas & Briden 1976; Piper *et al.* 1978; Claesson 1978). Since the Swedish limestone studied by Claesson is early Ordovician in age and if these steep magnetizations are valid records of an Ordovician dipole field, it suggests a large polar shift in the early Ordovician from around 30° S, 230° E to the vicinity of the Younger Gabbro and similar poles. This by no means exhausts the possibilities, however, for there are numerous quite different poles sited in the various papers mentioned above, for formations in Britain of presumed Ordovician age. At present there is no satisfactory explanation for the palaeomagnetic field in the Ordovician.

Consideration should also be given to the possibility that the field may not have been as strongly dipolar as now. In a survey of palaeointensity using basalt lavas and dykes, Carmichael (1967) showed the field in the Palaeozoic era was much weaker than either the Precambrian or the present field. Weak values for the field for various times in the Palaeozoic have been confirmed by several studies including an investigation of palaeointensity for part of the Devonian period by Kono (1979). This work supplements an extensive study of the Devonian lavas and sediments near Perth, Scotland, by Sallomy & Piper (1973). These investigations provide evidence that the field during this part of the Devonian was rather different from the Tertiary or recent field and Kono suggests that it may have had a higher proportion of non-dipole components. While there has not been a similar study of the Ordovician field, consideration should be given to evidence of the dipole nature of the field (well-defined magnetization directions with both polarities for instance) when assessing Ordovician and other Palaeozoic data. Thus, in addition to the technical problems of the resolution of multi-component magnetizations, tectonic relationships and precise determinations of the time of magnetization, the dipole–non-dipole nature of the field must be considered.

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